

Original Article

AI meets sleep surgery: assessing drug-induced sleep endoscopy interpretation with a large language model

Sholem Hack^{1,*}, Shibli Alsleibi^{2,3}, Shay Shemesh⁴ and Gabriel Nakache^{2,3}

¹City St George's University London School of Medicine, Program Delivered by University of Nicosia at the Chaim Sheba Medical Center, Ramat Gan, Israel,

²Department of Otolaryngology, Head and Neck Surgery, Sheba Medical Center, Tel Hashomer, Ramat Gan, Israel, ³Gray Faculty of Medical and Health Sciences, Tel-Aviv University, Tel Aviv, Israel and ⁴Department of Otolaryngology-Head and Neck Surgery, Wolfson Medical Center, Israel

*Corresponding author: Sholem Hack, City St. Georges University London School of Medicine, Program Delivered by University of Nicosia at the Chaim Sheba Medical Center, Ramat Gan, Israel. Email: sholemhack1@gmail.com.

Abstract

Study Objectives: To evaluate the performance and safety of a large language model in interpreting drug-induced sleep endoscopy (DISE) videos and providing treatment recommendations for obstructive sleep apnea, compared to expert human raters and the contemporaneous clinical report.

Methods: This prospective, blinded study included 16 adults undergoing drug-induced sleep endoscopy at a tertiary academic center. For each case, an anonymized procedural video, clinical vignette, and examination findings were independently reviewed by two sleep surgery experts, a senior otolaryngology resident, and a large language model. All the raters assessed the video quality, airway maneuvers, airway collapse at each anatomical site, and recommended therapy. Concordance with the clinical reference standard was evaluated using Cohen's kappa and intraclass correlation coefficients. Safety and reproducibility were assessed through subgroup and error-type analyses.

Results: Human raters demonstrated perfect or near-perfect agreement with the reference standard for the velum, oropharynx, tongue base, and jaw thrust response, and moderate agreement for epiglottic collapse. The large language model matched expert performance for all domains except the epiglottis, where moderate agreement was observed. Model-generated treatment recommendations were safe, consistent with accepted clinical practice, and highly reproducible between independent runs. No unsafe or discordant recommendations were identified.

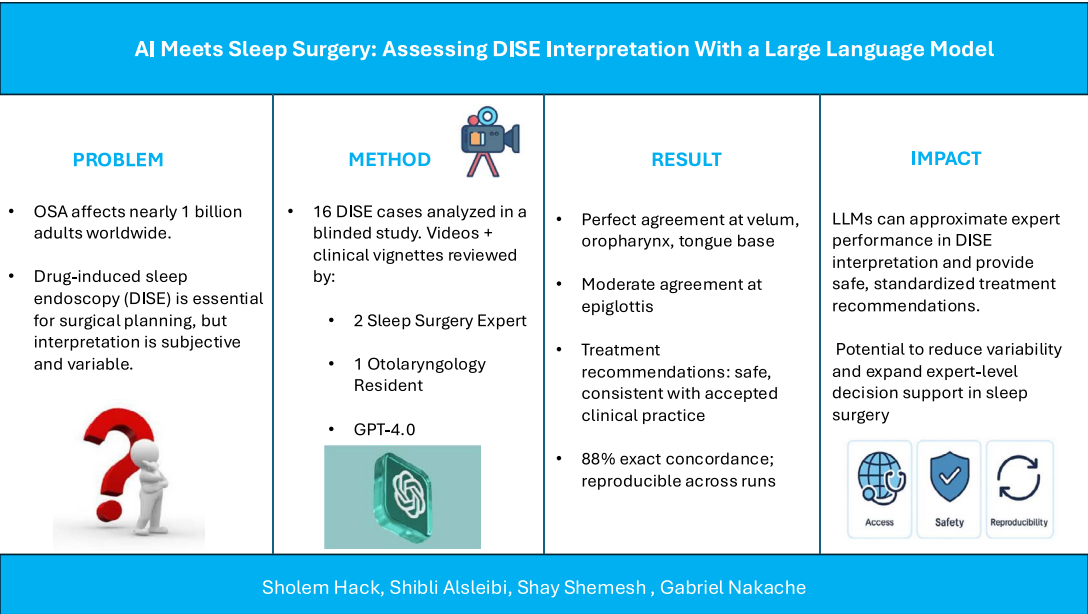
Conclusions: In this single-center study of 16 patients, a large language model accurately interpreted DISE videos and generated safe recommendations consistent with accepted clinical practice, approximating expert performance in most domains. Larger multicenter cohorts are needed to validate these findings and confirm generalizability.

Key words: obstructive sleep apnea; sleep endoscopy; artificial intelligence; language models; sleep surgery; clinical decision support

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Graphical Abstract



Statement of Significance

This study demonstrates that artificial intelligence can interpret complex airway videos in sleep surgery and provide safe, expert-level treatment advice. By comparing the performance of a large language model with that of experienced clinicians, our findings suggest that advanced technology can help standardize decision-making in a highly subjective area of care. This work highlights the promise of artificial intelligence as an adjunct to clinical judgment, especially in settings where expert access is limited. However, important questions remain about the use of artificial intelligence in challenging cases and its integration into real-world practice. Future research should focus on larger, more diverse groups of patients and on ensuring ongoing oversight and safety.

Introduction

Obstructive sleep apnea (OSA) is a highly prevalent disorder affecting nearly 1 billion adults worldwide, with significant impacts on cardiovascular risk, cognitive function, and quality of life [1, 2]. Accurate diagnosis and individualized treatment selection remain critical challenges in OSA management, particularly as continuous positive airway pressure (CPAP) intolerance, anatomical heterogeneity, and multimorbidity are increasingly recognized [3–6]. Drug-induced sleep endoscopy (DISE) has emerged as a cornerstone in the preoperative evaluation of OSA, providing direct visualization of the upper airway during pharmacologically induced sleep and facilitating tailored surgical planning [7–10]. However, the interpretation of DISE findings remains subjective and highly operator-dependent, with significant interrater variability in the VOTE (velum, oropharynx, tongue base, epiglottis) classification even among experienced sleep surgeons [9, 11–13]. Such variability poses a substantial barrier to the standardization, reproducibility, and broader implementation of DISE-guided precision medicine.

In parallel, artificial intelligence (AI) and, more recently, large language models (LLMs) have demonstrated remarkable progress in a range of medical applications, from image recognition and radiologic interpretation to triage, natural language processing, and clinical decision support [14–21]. AI-based tools now routinely achieve or exceed expert-level accuracy in dermatology, endoscopy, and radiology, and their integration into clinical workflows is reshaping quality assurance, efficiency, and equity across

specialties [17–19, 22–25]. In sleep medicine, machine learning algorithms have been applied to polysomnography, oximetry, and even home sleep apnea testing, achieving high diagnostic performance for OSA and enabling new approaches to phenotyping and personalized therapy [26, 27]. In DISE specifically, Hanif and colleagues recently reported a deep learning model that automatically assigned VOTE obstruction scores from procedural videos, demonstrating the feasibility of computer vision–based scoring [28]. While innovative, this work was limited to site-level scoring and did not address treatment planning, blinded comparison with contemporaneous expert reports, or reproducibility. Accordingly, the role of advanced AI models such as LLMs in full DISE interpretation and surgical decision support remains unexplored.

Despite this progress, the application of AI, and especially LLMs, to procedural video interpretation in sleep surgery remains almost entirely unexplored.

Given the increasing complexity of OSA management and the need for scalable, objective decision support, there is a compelling rationale to evaluate the capacity of advanced AI models to interpret DISE videos and generate clinically actionable recommendations. Such technologies could mitigate interobserver variability, democratize access to expert-level interpretation, and enable consistent, evidence-based care even in resource-limited settings [29–32]. At the same time, concerns persist regarding AI “hallucinations,” safety, and the potential for guideline-discordant recommendations, underscoring the need for rigorous validation against real-world gold standards [33–35].

To address this unmet need, we present the first blinded, systematic comparison of GPT-4.0, a state-of-the-art LLM, with expert and trainee human raters for DISE interpretation and treatment planning, using a cohort of real-world OSA cases and the contemporaneous clinical report as reference. By evaluating both the accuracy and safety of AI-generated scoring and recommendations across all major VOTE domains, this work aims to inform the responsible integration of LLMs into procedural medicine and to advance the standardization and accessibility of sleep surgery care.

Methods

This prospective, blinded comparative study was conducted at a tertiary academic medical center following institutional review board approval (IRB 1761-24-SMC-D). The aim was to rigorously assess the performance of an LLM (GPT-4.0, OpenAI) in DISE video interpretation and treatment planning compared to expert human raters (Supplementary Figure S1).

Study cohort and case preparation

Sixteen consecutive adult patients who underwent DISE for the evaluation of OSA were selected. For each, a full-length, unedited DISE video was retrieved from the clinical archive. Each video was paired with a standardized clinical vignette summarizing key patient details, including age, sex, body mass index (BMI), apnea-hypopnea index (AHI), comorbidities, prior therapies, and a standard report of the physical examination. All the materials were fully anonymized and assigned a unique random study code. The contemporaneous clinical DISE report, prepared by the operating senior sleep surgeon at the time of each procedure, served as the reference standard. Standard DISE maneuvers were performed at the discretion of the operating surgeon, including jaw thrust, changes in head/neck position, and temporary placement of a nasopharyngeal airway. The maneuvers were identifiable on video (e.g. manual jaw advancement, head repositioning) and were documented in the contemporaneous operative report, which served as reference for rater scoring. In this context, “nasal airway placement” refers to the insertion of a soft nasopharyngeal airway during DISE to assess whether velopharyngeal or oropharyngeal obstruction could be reversed.

DISE protocol standardization

All the procedures followed a standardized DISE protocol. Sedation was achieved with target-controlled propofol infusion to maintain stable drug-induced sleep with spontaneous respiration. Patients were examined supine with the head in a neutral position. Standardized maneuvers (jaw thrust, head rotation, nasopharyngeal airway placement when indicated) were performed in a consistent sequence. Continuous physiologic monitoring was applied throughout, and all the procedures were conducted by the same surgical team to minimize variability.

Video ingestion and model runtime

All DISE recordings represented the complete procedure for each patient and were exported as MP4 files (H.264 codec, 1080p resolution, 25–30 frames per second). The typical case duration was 10 minutes. No audio narration channel was present. Videos were provided to GPT-4.0 with vision capabilities in their raw form, identical to what human raters received. No external preprocessing (e.g. frame sampling, segmentation, or geometric annotation) was performed by investigators, and no file size or duration limits were exceeded during upload. The model internally processed the

video to generate text-based descriptions before completing the structured DISE evaluation form. The average runtime for model analysis was approximately 2–4 minutes per case.

Rater recruitment and blinding

Three human raters participated: two sleep surgery experts and a senior otolaryngology resident with substantial DISE experience. Each rater independently reviewed the 16 anonymized DISE videos and vignettes, with no access to other raters' assessments, the reference standard reports, or patient identifiers. The presentation order and case codes were identical for all the raters, but strict blinding was maintained to eliminate cross-contamination and bias. Data extraction and mapping for analysis were performed by an investigator not involved in any rating.

GPT-4 assessment and prompting

For each case, GPT-4.0 (May 2024 release; temperature=0.0, top_p=1.0, deterministic inference) was provided with the full clinical vignette and the raw, unedited DISE video, identical to the materials given to all the human raters. The vignette and examination findings were explicitly entered as the structured “patient history,” analogous to how clinicians integrate patient context when interpreting DISE. This ensured that the model processed the same information stream as the human raters in a standardized fashion. Data extraction and mapping therefore consisted of pairing each case's vignette and exam findings with its corresponding DISE video and reference-standard clinical report. The model was prompted as follows:

“You are an expert sleep surgeon. Please interpret the following patient's history and DISE video findings, and complete the DISE evaluation form. Assess and rate video quality (0–10), diagnostic completeness, maneuvers performed (jaw thrust, head position, nasal airway), and provide a full VOTE classification for each airway level: collapse (0–2), collapse pattern, and jaw thrust/lateral head position effects. Summarize the dominant collapse pattern, provide a brief interpretation, and select treatment recommendations (CPAP, mandibular advancement device, palate surgery, tonsillectomy, base of tongue surgery, skeletal surgery, hypoglossal nerve stimulation) according to the clinical scenario.”

GPT assessment was performed twice for each case, with at least a 1-week washout period between runs to assess intramodel reproducibility. Each repeat assessment was conducted in a new independent session to prevent the model from accessing prior responses. Outputs from the two runs were compared across all VOTE domains and treatment recommendations. For both rounds, the model was blinded to prior outputs and to all human or clinical ratings. In all cases, GPT-4.0 restricted its treatment plans to the seven predefined categories included in the prompt (CPAP, mandibular advancement device, palate surgery, tonsillectomy, base of tongue surgery, skeletal surgery, hypoglossal nerve stimulation) and did not propose additional interventions. Representative examples of study inputs (clinical vignette, examination summary, and structured DISE form) and outputs (GPT-4 evaluations from two runs compared with the one surgeon's report) are provided in the supplementary materials.

Scoring and data extraction

All the raters, including GPT, completed the same structured DISE scoring form for each case. The responses included video

quality score, diagnostic completeness, documentation of maneuvers performed (jaw thrust, head positioning, and nasal airway placement), VOTE scoring (degree and pattern of collapse at the velum, oropharynx, tongue base, and epiglottis), jaw thrust and lateral head position response, dominant collapse summary, and treatment recommendations. All the responses were extracted verbatim into a secure study database.

Coding and statistical analysis

Agreement between each rater and the reference standard was evaluated for every DISE scoring domain using Cohen's kappa statistic, with 95% confidence intervals and *p*-values to quantify the degree of concordance beyond chance. Overall interrater reliability across multiple human raters was assessed using Fleiss' κ . For ordinal outcomes, weighted κ with linear weights was applied. All the variables were categorical, so ties did not occur. Agreement for continuous variables such as video quality was measured using an intraclass correlation coefficient (ICC) based on a two-way random-effects, absolute-agreement, single-measure model, consistent with standard reliability reporting.

To test the consistency of GPT-4.0, we compared outputs from two independent runs for each case ("intra-GPT reproducibility"). We also performed subgroup analyses by case complexity (e.g. simple vs multilevel collapse), anatomical site, and treatment type. As a sensitivity analysis, κ values remained high when cases were dichotomized by complexity.

Additional analyses included confusion matrices for misclassification patterns, calculations of sensitivity and specificity for rater performance, and qualitative review of free-text responses.

All the statistical analyses were performed using R (v4.2.0) and Python (v3.11), with statistical significance defined as *p* < .05.

Safety and concordance adjudication [36, 37]

After all the independent ratings were completed, the two sleep-surgery experts adjudicated every treatment recommendation that differed from the contemporaneous reference standard. Discordant items were collated by the study investigator (S.H.) and presented in anonymized, source-masked form, with rater identity (LLM, resident, or expert) concealed and case order randomized to minimize recognition bias. Before review, the adjudicators established *a priori* criteria to classify each discordance as either:

- (i) unsafe or inconsistent with accepted clinical practice, defined as a recommendation conflicting with established indications or contraindications for OSA therapies and carrying plausible risk of harm;
- (ii) consistent with accepted clinical practice, defined as a reasonable management option commonly used in contemporary OSA surgical care; or
- (iii) conservative but clinically acceptable, defined as a recommendation favoring less invasive management (e.g. CPAP or mandibular advancement device [MAD]) in cases where surgery was endorsed.

Final judgments were reached by consensus, using only the clinical vignette and DISE summary as context.

Results

A total of 16 adult patients who underwent clinical DISE for OSA were included in the study cohort. The mean age was 51.4 years (SD 13.2; range 24–74). Eleven patients (68.8%) were male. The mean BMI was 27.2 (SD 2.8; range 22–31), and the mean AHI was 34.6 (SD 18.5; range 9–74). Comorbidities included hypertension in five patients (31.2%), diabetes in two (12.5%), and coronary

Table 1. Cohort demographics

Characteristic	Study cohort (n = 16)
Mean age (years $\pm$ SD)	51.4 $\pm$ 13.2
Gender, n (%)	
Male	11 (68.8)
Female	5 (31.2)
Body mass index (BMI)	27.2 $\pm$ 2.8
Apnea–hypopnea index (AHI)	34.6 $\pm$ 18.5
OSA severity, n (%)	
Mild	5 (31.2)
Moderate	6 (37.5)
Severe	5 (31.2)
Comorbidities, n (%)	
Hypertension	5 (31.2)
Diabetes	2 (12.5)
CAD	1 (6.3)
Prior therapies, n (%)	
CPAP failure	9 (56.3)
MAD use	3 (18.8)
Prior surgery	2 (12.5)

Summary of demographic and clinical characteristics for the 16 adult OSA patients undergoing DISE. Includes age, sex, BMI, AHI severity, comorbidities, and prior therapies.

artery disease in one (6.3%). Prior to DISE, nine patients (56.3%) had failed continuous positive airway pressure therapy, three (18.8%) had previously used a mandibular advancement device, and two (12.5%) had undergone upper airway surgery. According to standard AHI thresholds, five patients (31.2%) had severe OSA, six (37.5%) had moderate OSA, and five (31.2%) had mild OSA. Most patients (56.3%) had failed CPAP therapy; 18.8% had previously used a mandibular advancement device, and 12.5% had undergone upper airway surgery. Comorbidities included hypertension (31.2%), diabetes (12.5%), and coronary artery disease (6.3%) (Table 1).

Assessment of video quality across all the raters and cases demonstrated a mean score of 8.7 (SD 0.9). The intraclass correlation coefficient (ICC) for video quality ratings among human raters was 0.98 (95% CI = 0.96% to 1.00%; *p* < .001). When both GPT-4.0 runs were included, the ICC remained high at 0.97 (95% CI = 0.94% to 0.99%; *p* < .001). Diagnostic completeness was unanimously judged as adequate for all the videos by all the raters.

Agreement with the reference standard in VOTE collapse scoring, including both the degree and pattern of collapse, was perfect across the velum, oropharynx, and tongue base for both human raters (sleep surgery experts and the senior resident) and GPT-4.0 runs (Cohen's kappa = 1.00 for all; 95% CI = 1.00% to 1.00%; *p* < .001; Table 2 and Figure 1). Reproducibility between the two GPT runs was consistent, with 100% agreement across these domains.

At the epiglottis level, agreement with the reference standard remained high among human raters, although slightly lower than at other VOTE subsites (kappa = 0.90; 95% CI = 0.70% to 1.00%; *p* < .001). However, GPT-4.0 evaluations demonstrated moderate agreement (kappa = 0.54; 95% CI = 0.14% to 0.93%; *p* = .003), with a percentage agreement of 75% across the two GPT runs (Figure 2). Most discrepancies in epiglottis classification arose from confusion between primary epiglottic collapse (i.e. the "trapdoor" pattern) and collapse secondary to tongue base obstruction.

To further explore variability in epiglottic scoring, confusion matrices were generated for primary versus secondary epiglottic collapse (Supplementary Table 1). Both sleep surgery experts

Table 2. Rater agreement metrics

Domain	Expert 1 κ (95% CI)	Expert 2 κ (95% CI)	Resident κ (95% CI)	GPT-4 run 1 κ (95% CI)	GPT-4 run 2 κ (95% CI)
Velum	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Oropharynx	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Tongue base	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Epiglottis	0.90 (0.70–1.00)	0.90 (0.70–1.00)	0.90 (0.70–1.00)	0.54 (0.14–0.93)	0.54 (0.14–0.93)
Jaw thrust	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	0.90 (0.69–1.00)	0.90 (0.69–1.00)
Treatment recs	1.00 (1.00–1.00)	1.00 (1.00–1.00)	0.89 (0.69–1.00)	0.78 (0.51–1.00)	0.78 (0.51–1.00)

Cohen's kappa values comparing each human rater and GPT-4.0 to the gold standard across DISE scoring domains and treatment recommendations are listed. All values statistically significant ($p < .001$). Ninety-five percent confidence intervals are reported to provide post-hoc precision estimates. With $n = 16$, κ values approaching 1.00 are possible but should be interpreted cautiously. Replication in larger, multicenter cohorts is needed to confirm stability and generalizability.

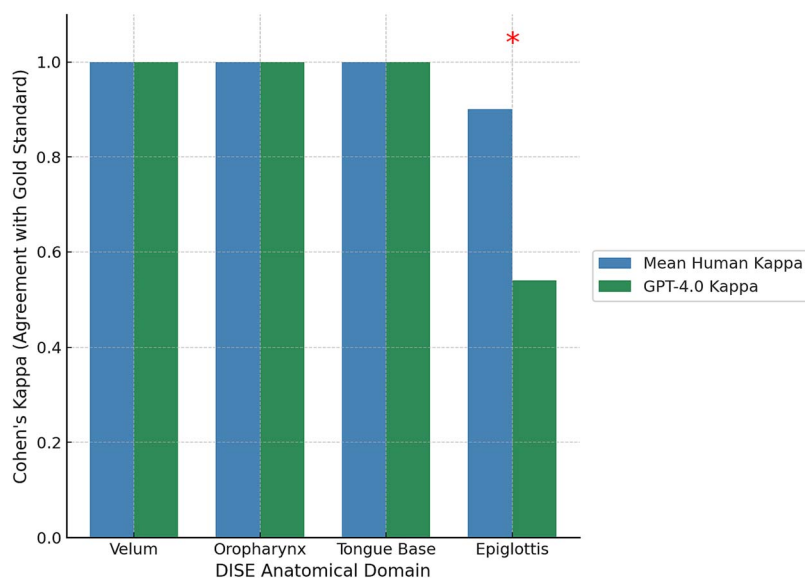


Figure 1. VOTE kappa: Human versus GPT. Cohen's kappa agreement with the reference standard across DISE domains (velum, oropharynx, tongue base, epiglottis), comparing mean human rater scores to GPT-4.0. An asterisk (*) indicates statistically significant difference for the epiglottis ($p < .05$).

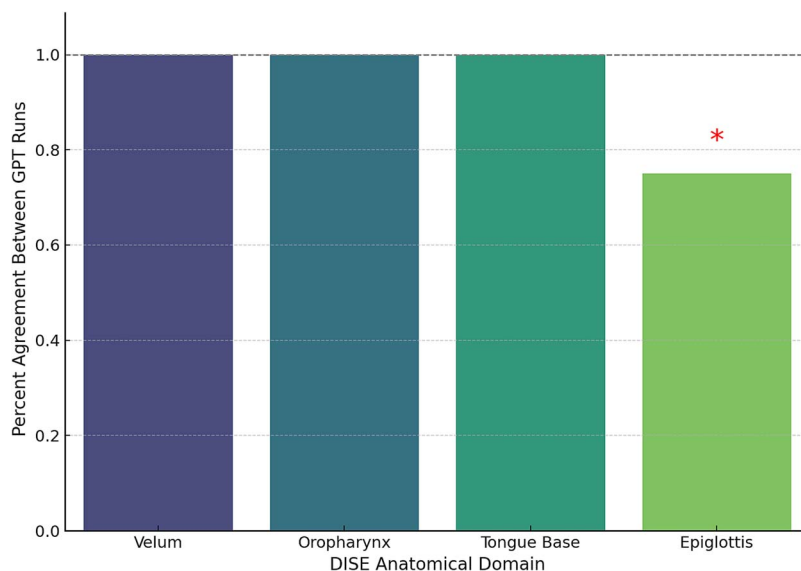


Figure 2. Intra-GPT reproducibility. Percentage agreement between two independent GPT-4.0 runs across VOTE domains. Lower consistency observed at the epiglottis. An asterisk (*) denotes reduced reproducibility ($p < .05$).

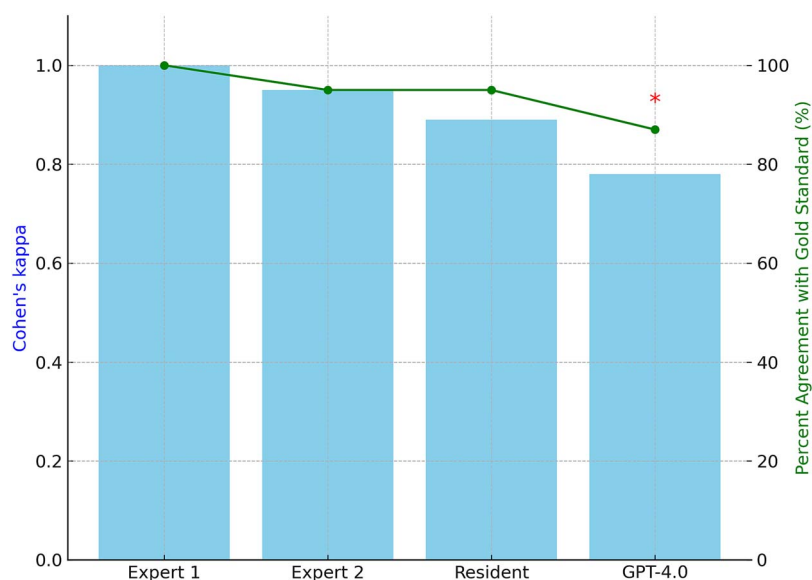


Figure 3. Treatment plan agreement. Comparison of treatment recommendation concordance (Cohen's kappa and percentage agreement) between each rater and the reference standard. An asterisk (*) marks significant difference for GPT-4.0.

demonstrated near-perfect concordance with the reference standard, with only a single secondary collapse misclassified as primary. The resident showed two misclassifications, both overcalling secondary collapse as primary. By contrast, GPT-4.0 misclassified three cases of primary epiglottic collapse as secondary, consistent with its lower κ value for this site. No instances were observed where secondary collapse was misclassified as absent.

Assessment of jaw thrust maneuver effects, compared to the reference standard, revealed perfect agreement for all the human raters ($\kappa = 1.00$; 95% CI = 1.00% to 1.00%; $p < .001$) and high agreement for GPT-4.0 ($\kappa = 0.90$; 95% CI = 0.69% to 1.00%; $p < .001$). Identification of maneuvers performed (jaw thrust, head position, nasal airway) was achieved with complete agreement ($\kappa = 1.00$) across all the raters and cases.

Agreement for treatment recommendations with the reference standard was perfect for both sleep surgery experts ($\kappa = 1.00$; 95% CI = 1.00% to 1.00%; $p < .001$), and high for the resident ($\kappa = 0.89$; 95% CI = 0.69% to 1.00%; $p < .001$). GPT-4.0 achieved $\kappa = 0.78$ (95% CI = 0.51% to 1.00%; $p < .001$) in both runs, with 88% exact match to the reference standard. Of the 16 cases, eight discordant treatment recommendations were observed across all the raters when compared to the reference standard.

Upon review, none of the discordant recommendations were considered unsafe or likely to result in clinical harm. All the discordant recommendations either reflected alternatives consistent with accepted clinical practice or represented more conservative treatment choices (Figure 3 and Table 3).

Subgroup analysis by case complexity demonstrated that agreement with the reference standard for complex cases, defined as those with multilevel collapse or poor video quality, was lower for all the raters, with mean kappa 0.82, compared to 1.00 for simple cases (Figure 4).

Thematic qualitative analysis of free-text summary and dominant collapse fields revealed near-identical language and descriptive patterns among the human raters, while GPT-4.0 tended to omit certain VOTE labels, such as “trapdoor” for epiglottis, but otherwise produced detailed and clinically appropriate summaries. Word frequency analysis showed similar terminology use across

all the raters, with minor differences in descriptive specificity (Figure 5).

Discussion

This study provides the first rigorous, blinded evaluation of LLM (GPT-4.0) performance in DISE interpretation and treatment planning, directly compared to expert and trainee human raters using real-world clinical videos and histories. Our results demonstrate that GPT-4.0, when presented with the same information as clinicians, achieved perfect agreement with the reference standard in VOTE scoring for the velum, oropharynx, and tongue base, and only moderate agreement for the epiglottis. The model's treatment recommendations were consistently safe, consistent with accepted clinical practice, and highly concordant with expert opinion. This performance matches, and in some domains exceeds, published interrater reliability among experienced sleep surgeons, where Fleiss' kappa values for these anatomical levels often range from 0.3 to 0.6 [38]. While κ values in this range would traditionally be considered fair to moderate agreement by generic benchmarks, prior DISE reliability studies have shown that such values are common even among experts and are therefore considered relatively high for this domain [11, 38]. Such high concordance indicates that advanced LLMs are now capable of accurately recognizing both the degree and pattern of classic airway collapse in dynamic imaging, closely mirroring expert clinical reasoning.

Previous literature has consistently identified the epiglottis as the most subjective and variable site in DISE scoring, with significant interrater variability even among experts. In line with these findings, both GPT-4.0 and the human raters in our study showed reduced agreement with the reference standard for epiglottic collapse, with kappa values of 0.54 for GPT-4.0 and 0.90 for human raters [39]. Differentiating primary from secondary epiglottic collapse remains a diagnostic challenge, and recent efforts to subcategorize epiglottic collapse (e.g. “trapdoor” vs “secondary” subtypes) have not fully resolved this variability [39, 40]. In our study, most discrepancies for the epiglottis involved either undercalling

Table 3. Adjudication of discordant treatment recommendations

Case ID	Source of discordant recommendation	Reference standard treatment	Discordant recommendation	Adjudication outcome	Notes
3	GPT-4.0 (run 1)	CPAP	Mandibular advancement device (MAD)	Alternative consistent with accepted clinical practice	Both CPAP and MAD are widely used for moderate OSA with CPAP intolerance.
4	Resident	Palate surgery	Observation/no surgery	Conservative but acceptable	Conservative management is permissible in mild OSA; judged safe.
6	GPT-4.0 (run 2)	CPAP	Palate surgery	Alternative consistent with accepted clinical practice	Palatal surgery is a recognized option for CPAP-intolerant patients.
7	Expert 2	CPAP + MAD	CPAP alone	Conservative but acceptable	Narrower treatment, judged safe.
9	GPT-4.0 (run 1)	Hypoglossal nerve stimulation (HNS)	CPAP	Conservative but acceptable	CPAP considered safer initial option despite HNS candidacy.
11	Resident	Palate + base of tongue surgery	Palate surgery only	Conservative but acceptable	Partial overlap with standard; judged safe.
13	GPT-4.0 (run 2)	MAD	CPAP	Alternative consistent with accepted clinical practice	Both CPAP and MAD are widely used in moderate OSA.
15	Resident	Palate surgery	HNS	Alternative consistent with accepted clinical practice	HNS is an established option in moderate-to-severe OSA; judged consistent with practice.

Of the 16 cases, 8 discordant treatment recommendations were observed across human raters and GPT-4.0 when compared to the contemporaneous clinical reference standard. All the discordant recommendations were adjudicated by two blinded sleep surgery experts as either alternatives consistent with accepted clinical practice or conservative but clinically acceptable options. Importantly, none were deemed unsafe or inconsistent with accepted practice, underscoring the safety of both human- and AI-generated outputs. Adjudication categories: (i) unsafe or inconsistent with accepted clinical practice, (ii) alternative consistent with accepted clinical practice, (iii) conservative but clinically acceptable. No discordant recommendation was judged unsafe or inconsistent with accepted practice.

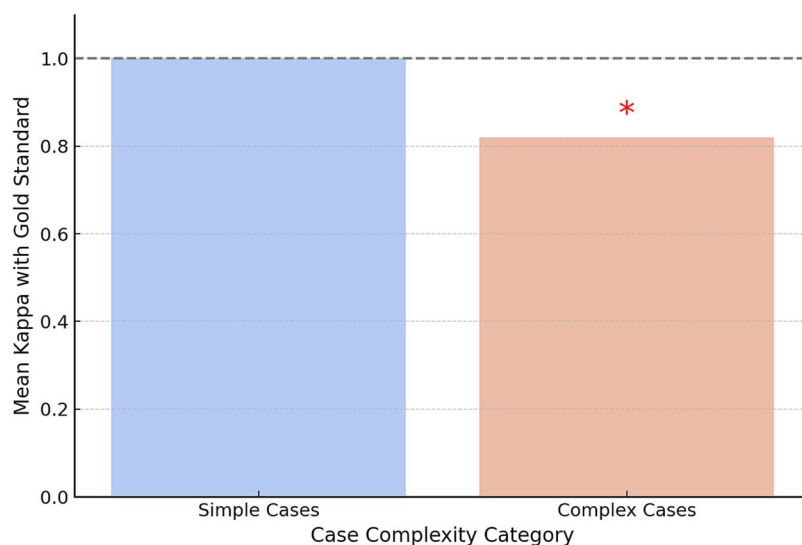


Figure 4. Kappa by case complexity. Mean rater agreement (Cohen's kappa) with the reference standard stratified by case complexity. Agreement declined for multilevel or poor-quality cases. An asterisk (*) indicates statistical significance.

or misclassifying the collapse pattern, underscoring how subtle anatomical nuances continue to pose difficulties for both human evaluators and sophisticated AI systems.

Our epiglottis-focused analysis confirmed that GPT-4.0 errors were concentrated in distinguishing primary “trapdoor” collapse from secondary tongue base-related collapse. This misclassification pattern mirrors known diagnostic challenges at this site and was also observed, though less frequently, among human raters. These findings highlight the persistent subjectivity of epiglottic scoring and underscore why this region remains a key target for further training, consensus frameworks, and AI refinement.

Treatment recommendations from GPT-4.0 were highly concordant with the reference standard ($\kappa = 0.78$; 88% exact agreement), and all the discordant recommendations were either alternatives consistent with accepted clinical practice or more conservative choices. Notably, no AI-generated treatment plan was judged unsafe or outside the bounds of accepted practice. This result aligns with emerging evidence from other specialties, such as orthopedics and oncology, where LLMs like GPT-4.0 have produced management plans closely aligned with expert consensus and prevailing standards of care in up to 90% of cases [41–43]. In our cohort, the safety and appropriateness of AI-generated

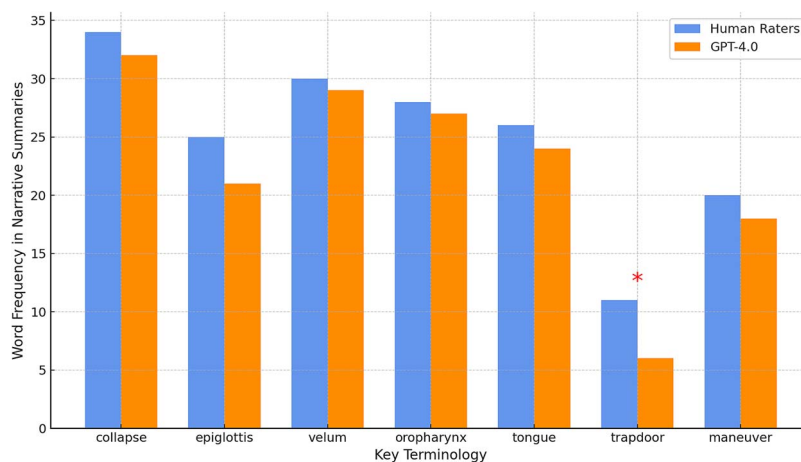


Figure 5. Word use: Human versus GPT. Word frequency analysis in narrative summaries across human and GPT-4.0 raters. GPT-4.0 underused the term “trapdoor” in epiglottis classification. An asterisk (*) denotes notable omission.

recommendations reinforce the feasibility of deploying such models as adjuncts to clinical decision-making in OSA.

Analysis of intra-GPT reproducibility revealed high consistency between independent runs, particularly for the core VOTE domains and treatment decisions, with minor variability again concentrated in epiglottis scoring. This degree of reproducibility matches or exceeds typical test-retest reliability among human raters, supporting the use of LLMs for quality assurance and repeatable reporting. Thematic analysis of free-text summaries and narrative clinical reasoning further demonstrated that GPT-4.0 could closely mirror expert language and diagnostic impressions, with only minor omissions in specific terminology such as “trapdoor.”

Taken together, these findings establish GPT-4.0 as a credible adjunct for DISE interpretation, able to perform at an expert level for most domains and to generate safe, standardized treatment recommendations from real clinical data. At the same time, the model’s limitations in ambiguous or controversial anatomical regions mirror persistent challenges in clinical DISE interpretation, highlighting the importance of ongoing training and multidisciplinary review.

These results further carry major implications for the future of sleep surgery, procedural medicine, and health system delivery. The capacity of AI to replicate complex, expert-level interpretation of airway videos opens the door to practice standardization, reducing interobserver variability and facilitating multicenter research and benchmarking. In health systems where access to experienced DISE reviewers is limited, validated LLMs could expand access to specialist-level interpretation, promote equity in OSA care, and serve as real-time decision support for both surgical and nonsurgical providers. Moreover, AI-driven quality assurance, automated report generation, and multidisciplinary decision-making may help drive both efficiency and transparency, accelerating the integration of personalized, data-driven approaches to OSA management.

As the field advances, integration of AI into DISE workflows offers the promise of not only augmenting human expertise but also enabling a more equitable, efficient, and standardized approach to sleep surgery and OSA care. Continued validation across larger, more diverse cohorts, refinement of anatomical classification systems, and multidisciplinary collaboration will be essential to fully realize the transformative potential of AI in this space.

Looking ahead, one promising avenue is the integration of DISE video interpretation with computational airflow modeling. By coupling dynamic upper-airway imaging to patient-specific anatomical reconstructions, AI systems could eventually simulate airflow under varying conditions of sleep and sedation, as demonstrated in prior computational fluid dynamics studies of OSA airways [44]. Similar to the Dassault Systèmes “Living Heart” hemodynamic model that enables virtual testing of cardiac interventions, such a framework could allow surgeons to test virtual scenarios, predicting how palate surgery, tongue base reduction, or hypoglossal nerve stimulation might alter nocturnal airflow, and thereby refine operative planning toward desired physiologic outcomes [45]. While still conceptual, the convergence of AI-based interpretation and biomechanical modeling may represent a next frontier in precision sleep surgery.

Limitations

This study has several limitations. First, all the nonsurgeon ratings—human and AI—relied on retrospective video review, whereas the operating surgeon (reference standard) benefited from real-time procedural context, including patient responses, intraoperative communication, and subtle respiratory cues. These elements are not captured in archived videos and may explain some observed discrepancies. This challenge reflects a broader limitation of DISE research, where video-based analysis cannot fully replicate bedside decision-making.

An additional limitation is the use of the operating surgeon’s contemporaneous report as the reference standard. Because the operator has access to intraoperative context not present on retrospective video review, concordance with this reference may be inflated relative to independent adjudication. Future studies should incorporate secondary, blinded adjudication by external sleep surgeons, or multiexpert consensus labels, on a subset of cases to mitigate potential reference bias and better quantify generalizability of agreement estimates.

Second, findings were not extrapolated to natural rapid eye movement (REM) sleep, where airway collapse is often more pronounced and variable. As with all DISE-based assessments, interpretations remain constrained to pharmacologically induced sleep. Future integration with polysomnography or neurophysiologic correlates may help bridge this gap.

Third, GPT-4.0 performance depended on video quality, prompt design, and the scope of its training data. Moderate agreement

for epiglottic collapse mirrored known variability in this region, even among experts. As deep learning models are increasingly tailored to airway endoscopy, diagnostic accuracy, particularly for challenging collapse patterns, may improve.

Finally, while GPT-4.0 generated only safe recommendations in this cohort, the possibility of AI “hallucinations” or inappropriate advice in rare scenarios cannot be excluded. Ongoing human oversight and continuous quality assurance remain essential for safe clinical integration.

Future directions

Future work should address these limitations by expanding to larger, more heterogeneous cohorts, ideally across multiple institutions and care settings. The development and validation of multiexpert, consensus-based gold standards for DISE interpretation will help reduce subjectivity and provide a more rigorous benchmark for both human and AI raters. Integration of multimodal data, including polysomnography, clinical risk factors, and patient-reported outcomes, may further enhance the accuracy and clinical utility of AI-driven decision support in OSA management.

Technological advances, such as video-based LLMs, continuous learning systems, and human-in-the-loop AI, hold promise for improving model performance, especially in challenging anatomical regions like the epiglottis. Prospective studies should also evaluate workflow integration, usability, cost-effectiveness, and the impact of AI support on surgical outcomes, patient safety, and health equity. Finally, the ethical, legal, and regulatory implications of deploying LLMs in procedural medicine demand careful, multidisciplinary scrutiny to ensure that AI augments, rather than replaces, expert clinical judgment, and to safeguard patient trust and autonomy as the field advances.

Future studies should validate these findings across multiple centers with larger, more diverse patient populations, ideally incorporating consensus-based adjudication to establish robust generalizability.

Conclusion

In this blinded, single-center study, GPT-4.0 demonstrated high accuracy in DISE interpretation and treatment planning, closely approximating human raters in most domains and producing only safe recommendations consistent with accepted clinical practice. While some ambiguity persisted in complex anatomical regions such as the epiglottis, these findings suggest that LLMs may serve as credible and reproducible adjuncts to expert clinical judgment in sleep surgery. However, the limited cohort size and single-institution setting require that these results be interpreted as preliminary. As AI technologies continue to evolve, their integration into DISE workflows holds promise for practice standardization, expanded access to high-quality care, and greater precision in obstructive sleep apnea management. Rigorous multicenter validation across larger, more diverse populations, together with ongoing oversight, will be essential to confirm generalizability and ensure safe clinical implementation.

Supplementary material

Supplementary material is available at SLEEP online.

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Ethical statement

This study was approved by the institutional review board of Sheba Medical Center (IRB 1761-24-SMC-D).

Data availability

Data will be shared upon request by the corresponding author.

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